LogiEdge Phase 1 Report

System architecture, hardware justification, and MQTT design

AIML ZG535 — Machine Learning on Edge - Continuous Evaluation Component EC-1

Birla Institute of Technology & Science, Pilani - Work Integrated Learning Programmes Division

Modules 1–3 (design). Measured training, quantisation, drift, and the deployment choice are in the Phase 2 report.

1 Constraint analysis

FreightBridge's pilot is 85 refrigerated trucks carrying medicines across Maharashtra, Karnataka and Andhra Pradesh. A refrigeration failure raises cargo temperature by about 1 °C per minute. Three minutes of an unseen fault moves a 4 °C shipment out of the Warning band, which is 1–3 °C from setpoint, and into Critical, which is more than 3 °C from setpoint. The system must detect that signature and raise an alert within 90 seconds of it appearing in the sensor data.

Cloud inference does not meet that bound on this network. A healthy rural round trip is about 50–500 ms, which would be fast enough to return a label if the link were up. The Nashik–Aurangabad route loses cellular coverage for 35–90 minutes at seven documented places. During those gaps a cloud pipeline receives no samples, so the round-trip clock never starts and no custody line is written. The edge node classifies on the truck. Building the 30-second feature window, stepped every 10 seconds, takes about 50 seconds in the worst alignment. Inference on the integer model is a fraction of a millisecond, so the alarm is raised locally and does not wait for a tower.

Each truck emits temperature at 1 Hz, three-axis vibration at 500 Hz, and door events. Temperature at 8 bytes a sample is 86,400 samples and 0.66 MB a day. Vibration at 12 bytes a sample is 43.2 million samples and 494 MB a day. One hundred door events add about 0.003 MB. The raw total is 494.66 MB per truck per day. At ₹0.10/MB that is ₹49.47 per truck per day, ₹4,204 a day for 85 trucks, and about ₹15.34 lakh a year, before any model update. The edge node uploads an alert only when the compartment state changes. Two hundred alerts of 200 bytes are about 0.04 MB a day per truck, a 99.99 percent cut, and the daily uplink cost falls to a fraction of a rupee.

A cloud-only truck in one of those gaps stops ingesting samples, misses a fault that is already warming the cargo, and cannot show a temperature chain of custody until the modem returns. The edge design keeps the broker, the model and the alert log on the vehicle. Each alert is written to disk and flushed before it is published. An outgoing bridge holds the queued alerts and delivers them in order when coverage returns, including after the edge node reboots.

Pharmaceutical clients require proof that cargo-condition data cannot be read by an unauthorised third party. On-device inference is that proof. Raw temperature history and the vibration stream never leave the truck. The broker binds to 127.0.0.1, so another device on the vehicle network cannot subscribe to the sensor topics. The cellular link carries only alerts and online/offline status, under mutual TLS with a certificate that can be revoked for one truck. The operations centre receives a classified event, not the sensor recording.

2. System architecture

Figure 1 is the truck. Temperature (1 Hz), vibration (500 Hz, three axes) and the door switch publish into a Mosquitto broker on the edge node. The Raspberry Pi 5 subscribes locally, filters, builds the six-feature window, runs the classifier, and publishes the class. A local log keeps every alert. A store-and-forward queue sends those alerts over cellular only when the link is up. The operations centre is a broker plus a dashboard. It is not in the inference path.

The drawing labels the local log as SQLite. The service writes alerts.jsonl on a Docker volume and fsyncs it before publish. That file is the same store-and-forward role: the record exists on the truck before any network send. Topic names in the figure are a short prefix. The implementation uses the tree in Section 4.

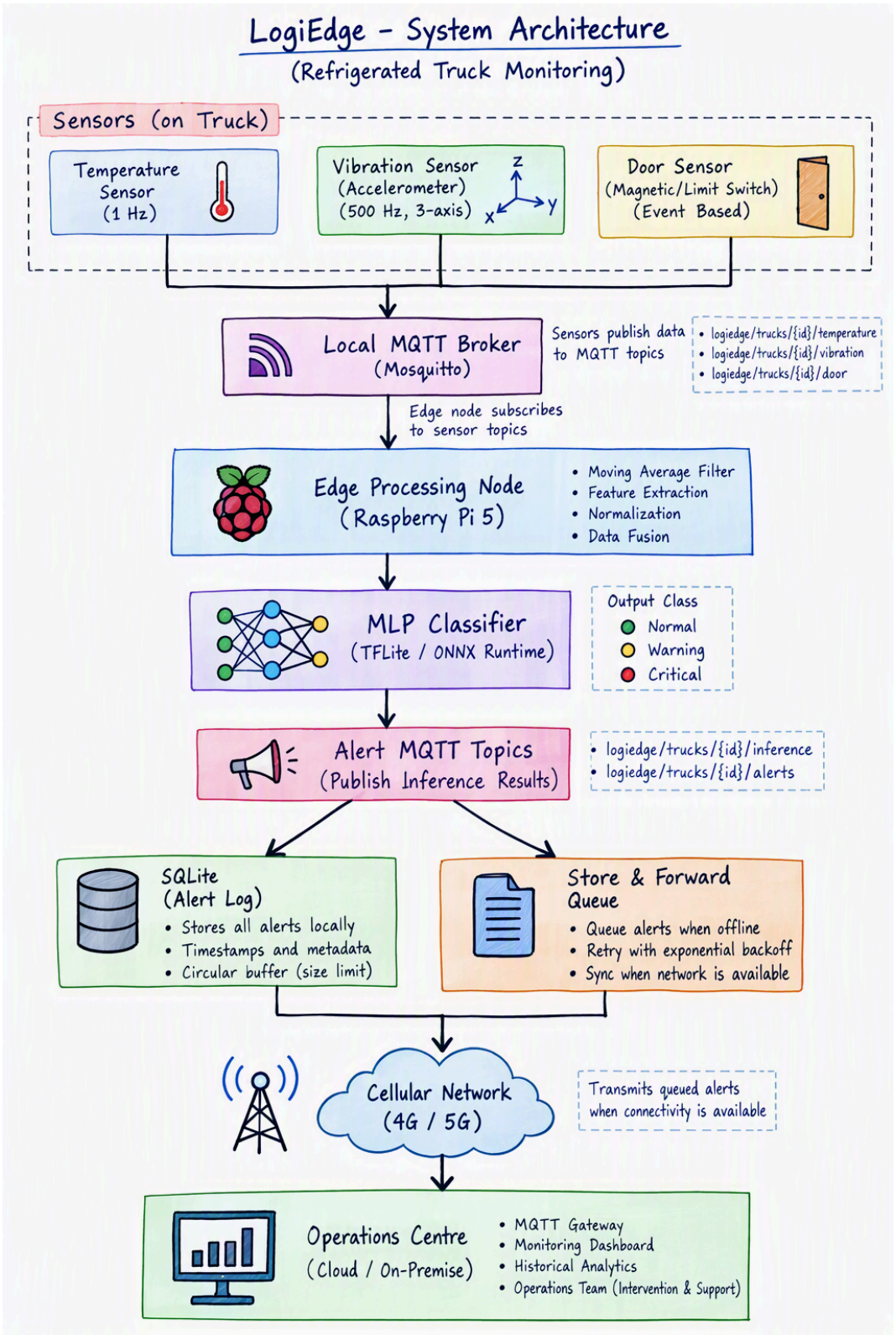

LogiEdge – System Architecture
(Refrigerated Truck Monitoring)
Sensors (on Truck)
Temperature Sensor (1 Hz)
Vibration Sensor (Accelerometer) (500 Hz, 3-axis)
z
y
x
Door Sensor (Magnetic/Limit Switch) (Event Based)
Local MQTT Broker (Mosquitto)
Sensors publish data to MQTT topics
• logiedge/trucks/{id}/temperature
• logiedge/trucks/{id}/vibration
• logiedge/trucks/{id}/door
Edge node subscribes to sensor topics
Edge Processing Node (Raspberry Pi 5)
• Moving Average Filter
• Feature Extraction
• Normalization
• Data Fusion
MLP Classifier (TFLite / ONNX Runtime)
Output Class
Normal
Warning
Critical
Alert MQTT Topics (Publish Inference Results)
• logiedge/trucks/{id}/inference
• logiedge/trucks/{id}/alerts
SQLite (Alert Log)
• Stores all alerts locally
• Timestamps and metadata
• Circular buffer (size limit)
Store & Forward Queue
• Queue alerts when offline
• Retry with exponential backoff
• Sync when network is available
Cellular Network (4G / 5G)
Transmits queued alerts when connectivity is available
Operations Centre (Cloud / On-Premise)
• MQTT Gateway
• Monitoring Dashboard
• Historical Analytics
• Operations Team (Intervention & Support)

Figure 1 caption: Sensors, the local broker, the Pi 5 pipeline, the on-board alert log, the cellular uplink and the operations centre. Inference does not leave the truck.

3. Hardware selection

Three boards were scored on the Constraint Triangle of latency, power and unit cost. Linux, Docker, MQTT and Ansible sit on the cost vertex, because a board that cannot take an update is expensive to own at 85 trucks and again at 265. The 10 W ceiling comes from the 12 V truck supply through a DC-DC converter. The 90-second alert bound is the latency test. Purchase price is the third vertex.

Board	Price	Power	10 W budget	Pilot (85)	Full fleet (265)
Pi 5 8 GB + Hailo AI HAT+	₹15,000	7.5 W	Inside	₹12.75 lakh	₹39.75 lakh
Jetson Orin Nano Super	₹45,000	15 W	Outside	₹38.25 lakh	₹1.19 crore
STM32H7 custom MCU	₹3,500	0.4 W	Inside	₹2.98 lakh	₹9.28 lakh

The Pi 5 is the recommendation. It meets the latency bound for this classifier, stays inside 7.5 W, and runs the container toolchain the rest of the project uses: local Mosquitto, a Docker inference service, and an Ansible playbook. The Hailo accelerator (about 13 TOPS) stays on the board for a later, heavier model. The network in this pilot has a few hundred parameters and runs on the Cortex-A76.

The Jetson Orin Nano Super is rejected. 67 TOPS is far above a six-input multilayer perceptron. 15 W misses the power budget. At ₹45,000 the pilot is ₹38.25 lakh and the full fleet is ₹1.19 crore, about three times the Pi 5, for compute this workload does not use.

The STM32H7 is rejected for the opposite reason. ₹3,500 and 0.4 W win on price and power, and the integer model would fit in its flash. Docker, Ansible and a repeatable TFLite update path are not practical on it. At 265 trucks that missing lifecycle costs more than the boards save. The dominant vertex is enough compute for the 90-second SLA, balanced against unit cost, with 10 W as a filter rather than the objective. The Pi 5 is the only option that passes the filter and can be operated as a fleet.

4. Arithmetic intensity and the Roofline

The brief's reference model does about 45 MFLOPs per inference and touches about 18 MB of weights and activations. The Pi 5 CPU is about 16 GFLOP/s with NEON, and the LPDDR4X bus is about 12 GB/s.

Arithmetic intensity is 45 million divided by 18 million, which is 2.5 FLOPs per byte. The ridge point is peak compute divided by peak bandwidth, 16 / 12 = 1.33 FLOPs per byte. The reference point sits above the ridge, so that workload is compute-bound. A faster memory bus would not move it. Fewer operations and cheaper arithmetic would: full INT8 quantisation, and structured pruning, which are the M2 and M3 variants in the Phase 2 report.

The network actually trained is much smaller than this reference, about 1.5 thousand FLOPs in FP32 and about 0.7 thousand after pruning, with a working set of 0.53 KB. It is still delivered as INT8 TensorFlow Lite and executed on the CPU. The Hailo part runs models compiled to its own format, not a `.tflite` file, and offloading a few hundred parameters would not change the 90-second budget.

5. MQTT design and data fusion

Every truck runs its own Mosquitto broker on 127.0.0.1:1883. The simulator, the inference service and, later, the drift monitor talk only to that broker, so classification continues with no cellular coverage. One outgoing bridge forwards alerts and status and queues them on disk while the link is down. Figure 2 is the tree the code publishes.

Figure 2 caption: Green topics leave the truck. Temperature, vibration, door events and the inference stream stay on the vehicle.

Topic	Rate	QoS	Leaves the truck?
truck/sensor/temperature	1 Hz	0	No
truck/sensor/vibration	0.5 Hz RMS	0	No
truck/sensor/door	per open or close	1	No
logiBridge/trucks/{id}/inference	every 10 s	0	No
logiBridge/trucks/{id}/alerts	on state change	1	Yes, queued
logiBridge/trucks/{id}/status	connect and last will	1, retained	Yes

Temperature and vibration use QoS 0. They are high-rate loopback streams, and a missed sample is OPEN or CLOSE corrupts the custody state, and a duplicate is harmless: the payload carries its timestamp. The inference topic uses QoS 0 and is not bridged. One lost window moves a histogram bin by about one percent, while bridging 8,640 messages a day would cancel the bandwidth saving in Section 1. Alerts use QoS 1, are persisted, and survive a reboot. QoS 2 would add a four-step handshake on a weak rural link. A sequence number lets the backend drop duplicates. Status is retained and has a last will, so a dead process is published as offline instead of looking like a healthy truck.

Preprocessing, specified here and measured in Phase 2, is a fixed sequence. A five-sample moving average feeds a 30-second window stepped every 10 seconds. The six features are temperature mean, standard deviation and rate of change in °C/min, plus vibration RMS, peak and kurtosis. Mean and standard deviation come from ten minutes of clean Normal data, saved as training_stats.npy, and are loaded at runtime. They are not recomputed from the live window.

Fusion is at feature level. Each sensor is reduced on its own, and the six numbers go to one classifier. Data-level fusion would resample 1 Hz against 0.5 Hz, and a 500 Hz accelerometer on the real truck, into one mixed-unit matrix much larger than a small edge model. Decision-level fusion, two classifiers combined by a vote, would drop the joint pattern this problem needs: temperature drift and a vibration step mean different classes, and only the combination separates Warning from Critical. One model on the concatenated vector keeps that interaction and stays small enough for the Pi 5.